

Differentiated Carbon Prices or International Transfers?

Adrien Fabre Constance Gorge

CNRS, CIRED — CIRED

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However **efficiency requires a uniform price**, and equity concerns can be settled by **transfers**.

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We compute the emissions rights equivalent to prominent differentiated prices proposals.

Differentiated prices *vs.* a uniform price

Why differentiate? Four imperfections

Absent imperfections, a uniform price with transfers is optimal (Aldy & Stavins, 2012). LICs allocated more rights than emissions: they clearly gain.

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⇒ Inconsistent with survey evidence (Fabre et al., 2025; Fabre, 2026).

None of the four decisively discard a uniform price: 1. uncalibrated, 2. irrelevant, 3. compatible, 4. unproven.

Standard Theory

The static case

Country i produces y_i , consumes c_i , and at carbon price p_i faces abatement cost a_i :

$$c_i = y_i - a_i + t_i, \quad t_i = \text{international transfer.}$$

Welfare u_i depends on c_i and on **global emissions** $E = \sum_j e_j n_j$, **fixed at E^***

n_i is population, e_i emissions p.c.

p^* : uniform price attaining E^* , i.e. **p^* s.t. $E^* = \sum_j e_j(p^*)n_j$.**

Shorthands $e_i^* := e_i(p^*)$, $a_i^* := a_i(e_i^*)$.

A **situation** for i is described by $s = (t_i, p_i)$. Two anchors:

Benchmark $s^* = (0, p^*)$: uniform price, no transfer.

Autarky $s^a = (0, p_i)$: differentiated price, no transfer.

u_i can be expressed as function of s : we have $u_i(0, p_i) = c_i^a = y_i - a_i$, and $u_i(t, p^*) = c_i^* + t = y_i - a_i^* + t$.

Step 1: the equivalent transfer

$V_i(p_i)$: the lump sum transfer which makes i as well off with the global price as in autarky:

$$u_i(V_i(p_i), p^*) = u_i(0, p_i) \quad \implies \quad V_i(p_i) = c_i^a - c_i^* = a_i^* - a_i.$$

Optimising country sets $\frac{da_i}{de}(e_i^*) = -p^*$: Therefore, a first-order Taylor expansion around e_i^* gives:

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Equivalent transfer g_i^s of an arbitrary situation $s = (t_i, p_i)$, relative to the benchmark:

$$g_i^s = \underbrace{t_i}_{\text{cash received}} + \underbrace{V_i(p_i)}_{\text{gain from facing } p_i \neq p^*}$$

Step 2: the correspondence between prices and rights

Cap-and-trade at p^* , each person in i endowed with rights r_i . Net monetary transfer to i :

$$t_i^* = (r_i - e_i^*) p^* \quad \implies \quad g_i^* = t_i^*.$$

Autarky, from Step 1: $g_i^a = V_i(p_i) \approx (p^* - p_i) e_i^* \varepsilon^*$. Setting $g_i^* = g_i^a$:

$$\begin{aligned} & \boxed{(r_i - e_i^*) p^* \approx (p^* - p_i) e_i^* \varepsilon^*} \\ \iff & \quad r_i \approx \left(1 + \left(1 - \frac{p_i}{p^*}\right) \varepsilon^*\right) e_i^* \quad \iff \quad p_i \approx \left(1 + \frac{e_i^* - r_i}{e_i^* \varepsilon^*}\right) p^*. \end{aligned}$$

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The key formula for policy design: rights r_i under uniform price equivalent to autarky price p_i

Low autarky price ($p_i < p^*$) $\Rightarrow$ more rights than emissions $r_i > e_i^*$ to compensate the extra abatement cost.

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$\Rightarrow$ Any differentiated-price arrangement maps onto a cap-and-trade allocation.

If the two can always be made equivalent, *is there any reason to prefer one?* Yes.

Step 3: the uniform price weakly dominates, for *every* country

Give country i rights equal to its *autarky* emissions, $r_i = e_i$. Net transfer $\tilde{t}_i = (e_i - e_i^*)p^*$.

Convexity of abatement costs implies:

$$a_i - a_i^* \geq (e_i - e_i^*) \frac{da_i^*}{de} = -(e_i - e_i^*)p^* \quad \implies \quad \tilde{t}_i \geq a_i^* - a_i = V_i(p_i) = g_i^a.$$

Hence, consumption increases *in every country* when trading is allowed and prices equalize:

$$\tilde{c}_i = y_i + \tilde{t}_i - a_i^* \geq y_i - a_i = c_i^a$$

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This is [Diamond & Mirrlees \(1971\)](#) production efficiency applied to climate: absent other distortions, **direct transfers beat differentiated factor prices as a redistributive instrument.**

The dynamic case: the equivalence breaks

Add time t ; each variable becomes a trajectory. β_t = discount factor, $R_i = \sum_t r_{it} = i$'s carbon budget.

$G_i = \sum_t g_{it} \beta_t$: total discounted equivalent gain

Welfare equivalence now requires equality of *entire discounted streams*, $G_i^U = G_i^A$:

$$\sum_t \beta_t (r_{it} - e_{it}^*) p_t^* \approx \sum_t \beta_t (p_t^* - p_{it}) e_{it}^* \varepsilon_t^*.$$

In the static case p^* was a *common factor* and cancelled. Here it does not: the timing of allocation matters.

A **generous** allocation in a **low-price** period is **worth less** than the same allocation when prices are high.

⇒ **No direct correspondence between the intertemporal carbon budget R_i and welfare.**

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Special case 1 — homothetic prices, fixed emission rights shares: $p_{it} = \pi_i p_t^*$ and $r_{it} = \rho_i r_t$

$$\mathcal{B} := \sum_t r_t \beta_t p_t^* \quad \rho_i \mathcal{B} - \sum_t e_{it}^* p_t^* \beta_t \approx (1 - \pi_i) \sum_t e_{it}^* \varepsilon_t^* p_t^* \beta_t,$$

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Analogous to static formula, but $\mathcal{B}$ still requires the full price path $\Rightarrow$ **not tractable without a model.**

Special case 2: Hotelling — and why it is the right benchmark

Hotelling rule: Fixed aggregate carbon budget + perfect foresight + efficient capital markets
 $\Rightarrow$ carbon price rises at the rate of return on capital. Discounting at that same rate, $\beta_t p_t^* = p_0$ is constant.

$\beta_t p_t^*$ then factors out of both sums, and the dynamic condition collapses to:

$$\boxed{R_i - E_i^* \approx (1 - \pi_i) \sum_t e_{it}^* \varepsilon_t^*} \quad E_i^* = \sum_t e_{it}^*$$

Structurally identical to the static formula: the intertemporal carbon budget R_i plays the role of the single-period allocation, and the homothetic price factor π_i the role of p_i/p^* .

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In the numerical analysis, we place ourselves in the Hotelling case.

Numerical analysis

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Grid search over (ρ_i, π_i) for eight major regions: **USA, Russia, EU27, China, Turkey, India, Nigeria, DRC**.

The two scenarios

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Revenue recycled domestically under Negishi weighting (transfers $\propto$ inverse marginal utility)
 $\Rightarrow$ neutralises the domestic redistribution component, **isolating the pure cross-country effect**.

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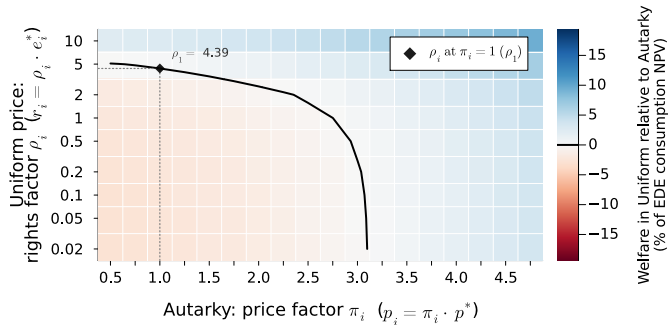
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RoW receives the *residual* cap, grandfathered on baseline emissions.

Reading the heatmaps

USA welfare **gain of uniform over autarky**, depending on rights and price.

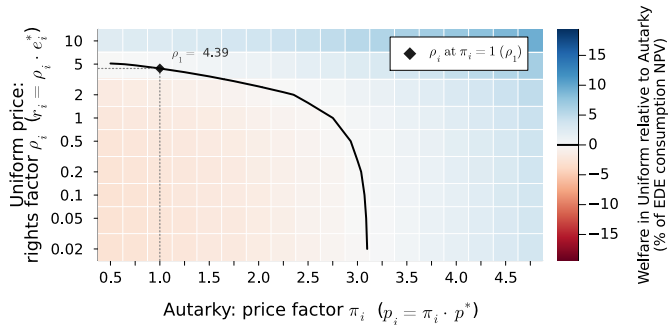


Grid showing the welfare **gain** or **loss** of *Uniform over Autarky*

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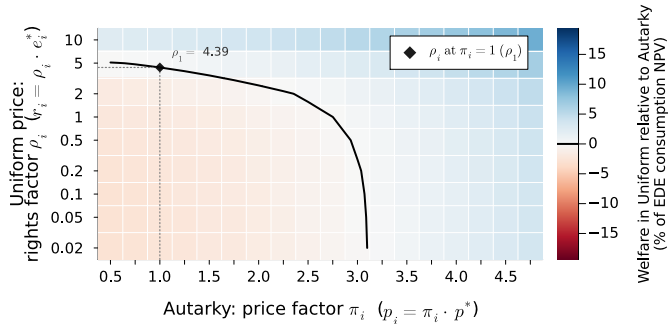
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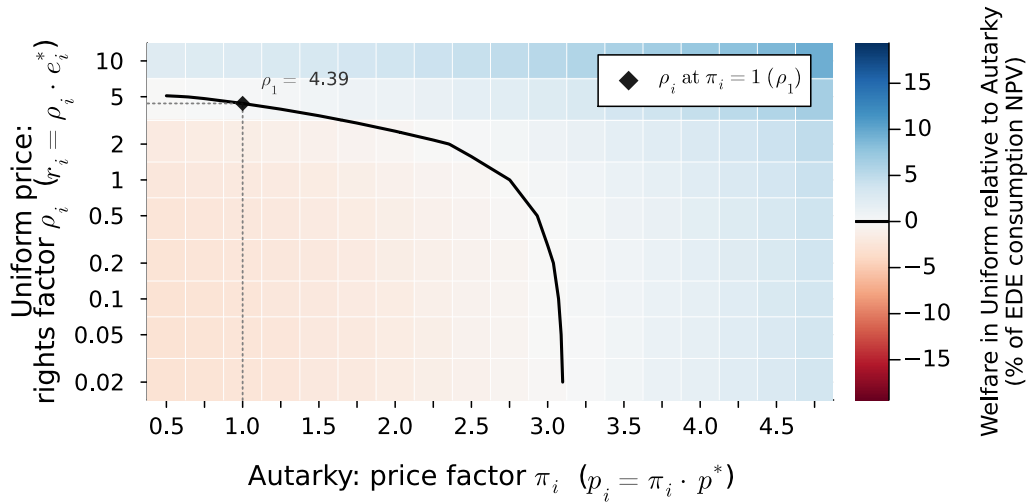
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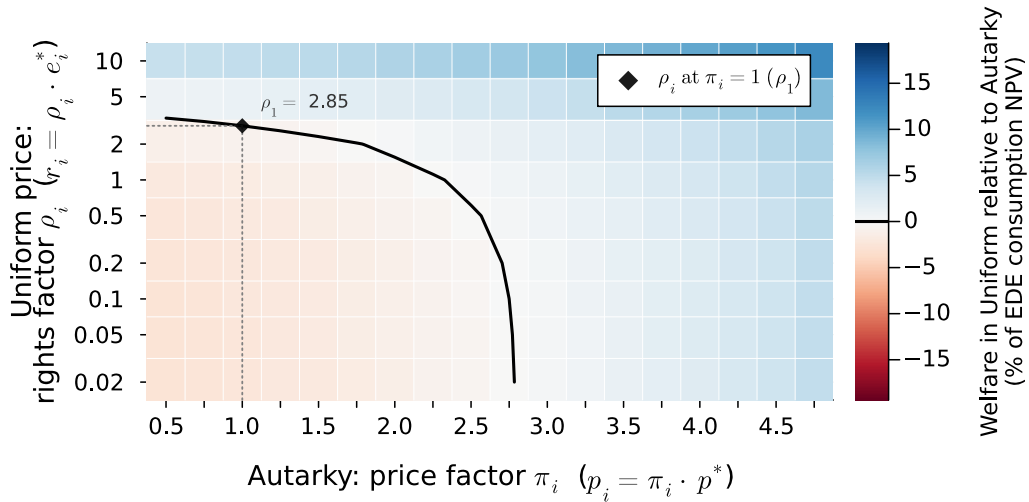
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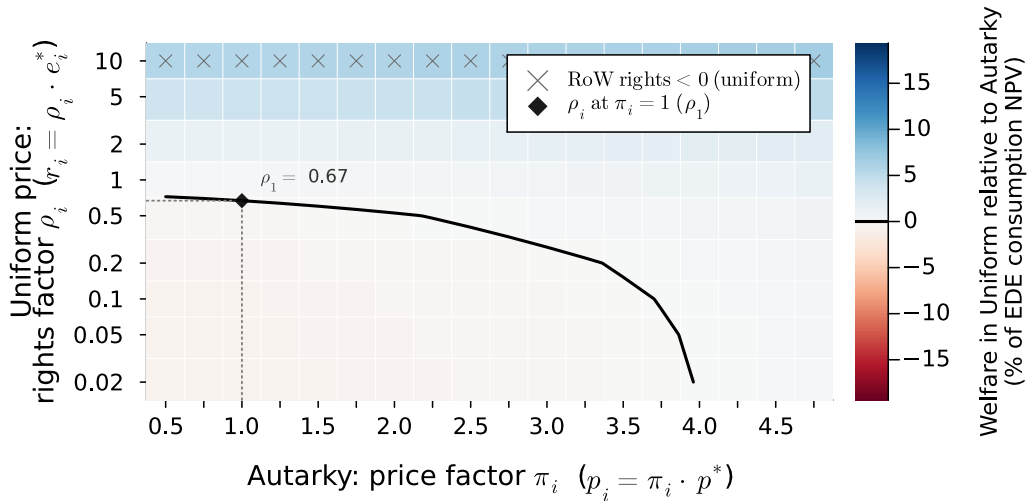
United States



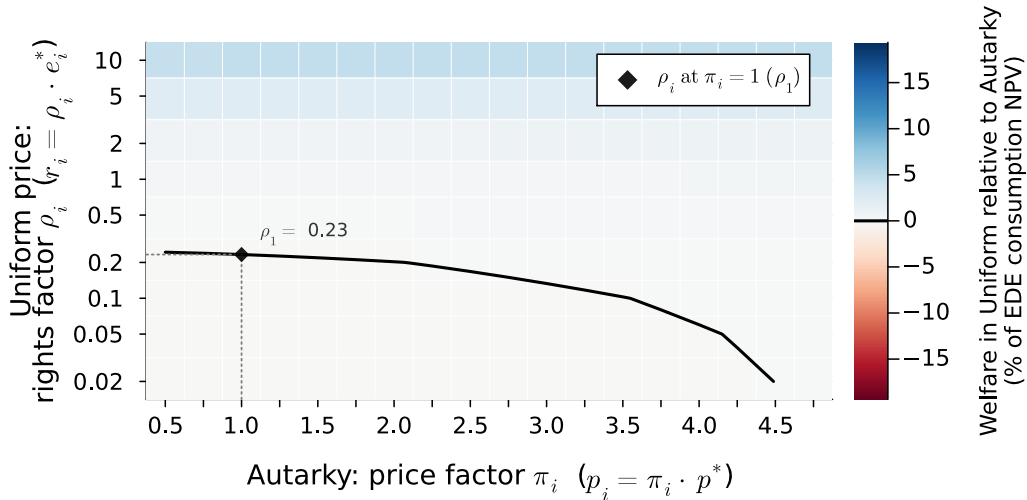
Russia



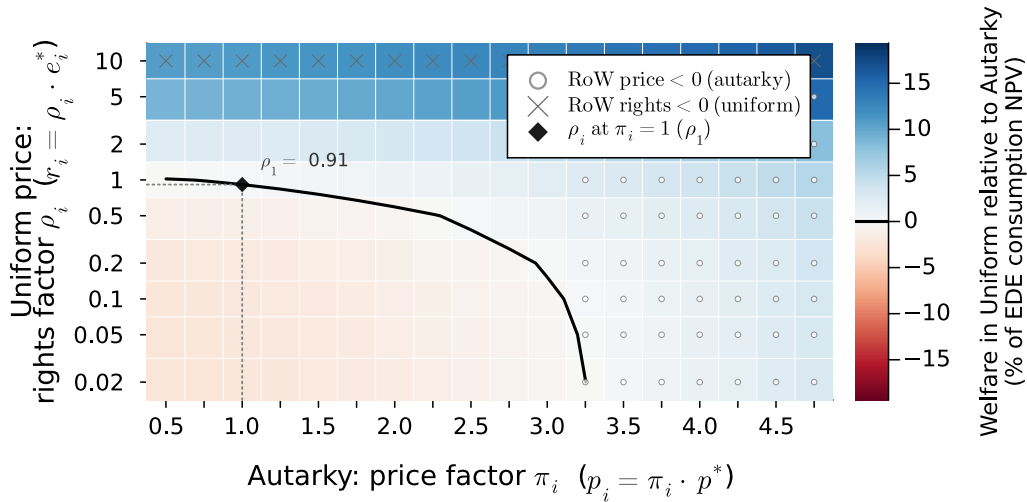
China



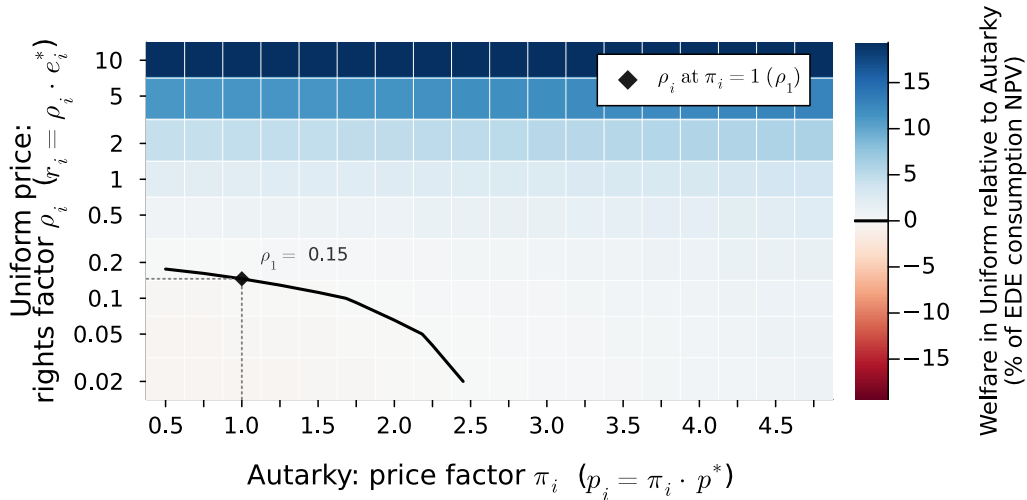
European Union (EU27)



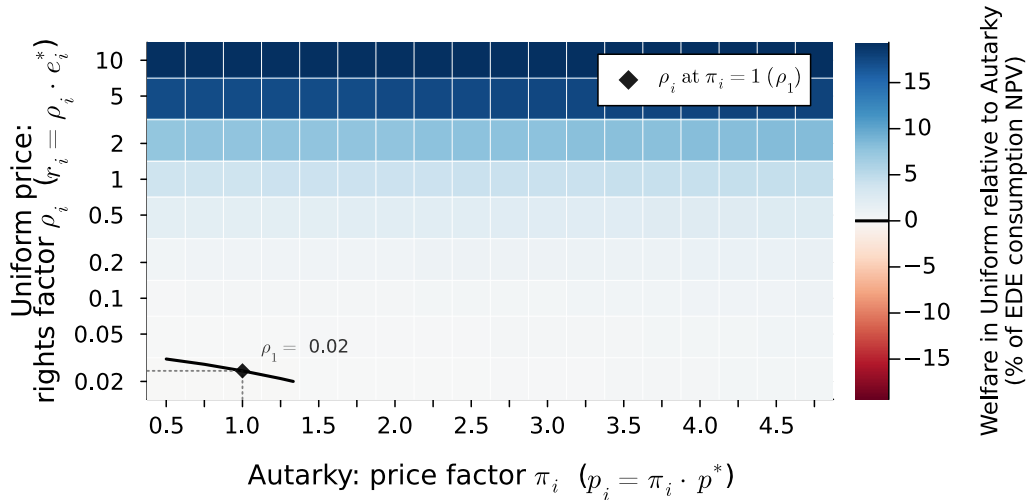
India



Nigeria



Democratic Republic of the Congo



Assessment of live proposals

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We price HICs at \$75/t (Wolfram's top tier): unpriced HICs would make the “bargain” a cut in world

Equivalent rights for the proposals

Country	Wolfram et al.			Banerjee et al.		
	p_i	ρ^A	ρ^B	p_i	ρ^A	ρ^B
United States	0	—	—	75	2.16	2.82
Russia	0	—	—	75	1.36	1.89
European Union	75	0.26	0.36	75	0.26	0.36
Turkey	0	—	—	50	1.54	1.62
Indonesia	50	0.89	0.92	50	1.09	1.04
China	50	0.84	0.98	50	0.88	0.95
Brazil	50	0.38	0.37	50	0.5	0.41
India	25	1.3	1.12	30	1.18	1.03
Nigeria	0	—	—	30	0.24	0.11
DR Congo	0	—	—	10	0.06	0.01
Gain in rights (variant 1)		9.9%	9.5%		9.1%	8.5%
World temp. 2100, change (°C)		−0.020	−0.019		−0.037	−0.034
World welfare gain, EDE (variant 2)		0.028%	−0.006%		0.102%	−0.009%
World consumption gain (variant 2)		0.022%	0.018%		0.057%	0.048%
World welfare gain, EDE (variant 3)		0.028%	0.011%		0.102%	0.041%
World consumption gain (variant 3)		0.022%	0.020%		0.057%	0.047%
World welfare gain, EDE (variant 4)		0.081%	0.048%		0.419%	0.298%
World consumption gain (variant 4)		0.024%	0.021%		0.068%	0.059%

The comparison regime prices the proposal's own members at a single price (\$49.2/t Wolfram et al., \$55.8/t Banerjee et al. in 2030) and leaves non-members unpriced, as the proposal does. p_i : the price the proposal asks of i in 2030 (\$/t); a country it does not price is shown at $p_i = 0$ with no equivalent allocation ('—'). ρ_i : allocation as a multiple of an equal-per-capita share of the club's emissions; ρ^A solves each country alone, ρ^B all jointly (distribution solved to 0.03–0.05%, level held at option A's). Variant 1 lets the rights set the cap, so the temperature row is the change from the proposal's own 2100 warming (2.16°C Wolfram et al., 1.80°C Banerjee et

Conclusion

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A **first-best statement**. The dominance result assumes away every other distortion. Where the second-best arguments bite — market power, pre-existing sectoral taxes, divergent discount rates (Anthoff et al., 2022; Babiker et al., 2004) — differentiation can be optimal in its own right, and we are **silent**. ▶ The three arguments

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The dynamic equivalence leans on Hotelling. Imperfect foresight or frictional capital markets break the constancy of $\beta_t p_t^*$ and **reopen the timing problem** the static formula cannot see. ▶ Why Hotelling

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Thank you for your attention!

adrien.fabre@cnrs.fr

constance.gorge@cired.fr

Appendix

Two boundary conditions evaluated in *every* year of 2030–2100:

1. **Market-distorting subsidies** ($\omega_i \times \pi_i > 1$): the target country claims more than 100% of the global price burden. To compensate, the model is forced to push $p_{-i} < 0$ — economically, a **fossil-fuel subsidy for the RoW**.

2. **Negative emission rights** ($\rho_i > 1 / pop_i$): granting i its multiplier uses more than the whole global budget, forcing the RoW into **negative rights**. Binds for large populations: countries above 10% of world population (China, India) trigger it for any $\rho_i > 10$.

RoW price identity, the starting point of the root-finding:

$$p^* = \omega_i p_i + (1 - \omega_i) p_{-i}, \quad \omega_i = e_i / E^* \quad \implies \quad p_{-i} = p^* (E^* - e_i \pi_i) / e_{-i}.$$

This linear approximation assumes emission-weighted price averaging delivers the same global abatement — false with non-linear MACs, hence the **Newton–Raphson** correction to $< 0.1\%$ relative error on the cap.

Autarky scenario, United States ($\pi = 2$) [Go back](#)

Variable	2030	2040	2050	2060	2070	2080	2090	2100
p^* (USD/tCO ₂)	42.6	112.9	205.9	285.7	401.6	480.9	476.2	471.4
p_i (USD/tCO ₂)	85.2	225.8	411.7	571.4	803.3	961.9	952.3	942.8
E_i (GtCO ₂)	4.31	2.58	0.63	-0.53	-1.75	-2.22	-1.86	-1.52

Net-negative US emissions from the 2060s: abatement rates $\mu_i > 1$, i.e. [CDR deployment](#), with the MAC function extrapolated beyond the 100% boundary.

Standard IAMs constrain $\mu_i \in [0, 1]$; enforcing that bound crashes the grid search under extreme policy parameters.

The equivalent price, and why we do not use it [▶ Go back](#)

One could instead define an **equivalent price** p_i^* such that $(0, p_i^*)$ is welfare-equivalent to (t_i, p_i) :

$$t_i + (p^* - p_i)e_i^*\varepsilon^* \approx (p^* - p_i^*)e_i^*\varepsilon^* \quad \implies \quad p_i^* \approx p_i - \frac{t_i}{e_i^*\varepsilon^*}.$$

Problem: p_i^* can be *negative* when the transfer is large, which has no natural interpretation as a carbon price.

The equivalent transfer g_i^s does not share this limitation $\implies$ we use it as the primary welfare metric.

Last slide (needed)